

Q1. What is a database?

A. A database is an organized collection of data that can be stored, managed, and retrieved efficiently. Examples include MySQL, PostgreSQL, Oracle, and MongoDB.

Q2. What is DBMS?

A. A Database Management System (DBMS) is software that enables users to create, manage, retrieve, and secure data in databases. Examples include MySQL, PostgreSQL, Oracle, and SQL Server.

Q3. SQL vs NoSQL?

A. SQL databases use structured tables and predefined schemas. NoSQL databases use flexible schemas and are designed for scalability and unstructured data.

Q4. What is Normalization?

A. Normalization is the process of organizing data to reduce redundancy and improve data integrity.

Q5. What is Denormalization?

A. Denormalization intentionally introduces redundancy to improve read performance and query speed.

Q6. What is a Primary Key?

A. A primary key uniquely identifies each row in a table and cannot contain NULL values.

Q7. What is a Foreign Key?

A. A foreign key is a field that establishes a relationship between two tables by referencing the primary key of another table.

Q8. What are Indexes?

A. Indexes improve query performance by allowing the database to locate rows more efficiently.

Q9. Clustered vs Non-Clustered Indexes?

A. A clustered index determines the physical order of data in a table. A non-clustered index stores pointers to the actual data.

Q10. What are Views?

A. Views are virtual tables created from SQL queries that simplify data access and improve security.

Q11. What are Stored Procedures?

A. Stored procedures are precompiled SQL programs stored in the database and executed when called.

Q12. What are Triggers?

A. Triggers are automatically executed actions that occur when INSERT, UPDATE, or DELETE events happen.

Q13. What is a Transaction?

A. A transaction is a sequence of operations treated as a single unit of work.

Q14. What is ACID?

A. ACID stands for Atomicity, Consistency, Isolation, and Durability. These properties ensure reliable transaction processing.

Q15. What are Isolation Levels?

A. Isolation levels control how transactions interact and prevent issues such as dirty reads and phantom reads.

Q16. What are Deadlocks?

A. Deadlocks occur when two or more transactions wait indefinitely for resources held by each other.

Q17. What is Replication?

A. Replication copies data from one database server to another to improve availability and fault tolerance.

Q18. What is Sharding?

A. Sharding distributes data across multiple servers to improve scalability and performance.

Q19. What is Partitioning?

A. Partitioning divides large tables into smaller, manageable pieces while maintaining logical consistency.

Q20. What are Backup Strategies?

A. Backup strategies include full, differential, and incremental backups to ensure data recovery.

Q21. What is Disaster Recovery?

A. Disaster recovery is the process of restoring database operations after failures, disasters, or outages.

Q22. What is Failover?

A. Failover automatically switches operations to a backup system when the primary system fails.

Q23. What is Query Optimization?

A. Query optimization improves SQL query performance through indexing, rewriting queries, and analyzing execution plans.

Q24. What is an Execution Plan?

A. An execution plan shows how the database engine processes a query and accesses data.

Q25. What is Database Tuning?

A. Database tuning involves optimizing configuration, queries, indexes, and hardware for better performance.

Q26. What is Locking?

A. Locking controls concurrent access to data and prevents conflicts between transactions.

Q27. What is Concurrency Control?

A. Concurrency control ensures multiple users can access data simultaneously without compromising consistency.

Q28. What is Data Integrity?

A. Data integrity ensures data remains accurate, consistent, and reliable throughout its lifecycle.

Q29. What is ETL?

A. ETL stands for Extract, Transform, and Load, a process used to move and prepare data for analysis.

Q30. What is Data Warehousing?

A. A data warehouse is a centralized repository designed for reporting, analytics, and business intelligence.

Q31. What is OLTP?

A. Online Transaction Processing (OLTP) systems handle high volumes of real-time transactional operations.

Q32. What is OLAP?

A. Online Analytical Processing (OLAP) systems support complex analytical queries and reporting.

Q33. What is MongoDB?

A. MongoDB is a NoSQL document database that stores data in flexible JSON-like documents.

Q34. What is PostgreSQL?

A. PostgreSQL is an open-source relational database known for reliability, extensibility, and standards compliance.

Q35. What is MySQL?

A. MySQL is a popular open-source relational database widely used in web applications.

Q36. What is Oracle DB?

A. Oracle Database is an enterprise-grade relational database management system developed by Oracle Corporation.

Q37. What is Redis?

A. Redis is an in-memory data store commonly used for caching, messaging, and real-time applications.

Q38. What is Caching?

A. Caching stores frequently accessed data in fast memory to improve application performance.

Q39. What is Database Security?

A. Database security protects data from unauthorized access, misuse, and cyber threats.

Q40. What is Role-Based Access Control?

A. RBAC restricts system access based on user roles and assigned permissions.

Q41. What is Encryption at Rest?

A. Encryption at rest protects stored data by encrypting it on disks or storage devices.

Q42. What is Encryption in Transit?

A. Encryption in transit protects data while it is being transmitted across networks.

Q43. What is a Schema?

A. A schema defines the structure of a database, including tables, relationships, and constraints.

Q44. What is a Migration Strategy?

A. A migration strategy outlines how database changes are deployed while minimizing downtime and risk.

Q45. What is Database Monitoring?

A. Database monitoring tracks performance, availability, resource usage, and potential issues.

Q46. Tools Used?

A. Common DBA tools include pgAdmin, MySQL Workbench, Oracle Enterprise Manager, SQL Server Management Studio, and monitoring solutions.

Q47. Performance Issue Example

A. Describe the issue, investigation process, root cause analysis, optimization steps, and results achieved.

Q48. Recovery Scenario

A. Explain how backups, replication, or disaster recovery procedures were used to restore services.

Q49. Handling Downtime

A. Minimize downtime through failover mechanisms, maintenance planning, monitoring, and rapid incident response.

Q50. Questions for Us?

A. Ask about database technologies, team structure, scalability challenges, disaster recovery processes, and growth opportunities.